Alessio Olivieri

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github.com/Alessio-Olivieri | linkedin.com/in/alessio-olivieri | Portfolio

EDUCATION

MSc, Artificial Intelligence & Robotics Jan 2025 – May 2027 (expected)
Sapienza University of Rome | Average: **29/30**.

BSc, Applied Computer Science & Artificial Intelligence Oct 2020 – Dec 2024
Sapienza University of Rome | Final grade: **101/110**.
Thesis (2024): Enabled Swing AllReduce on GPUs with MSCCLang/NCCL; benchmarked on Leonardo.

EXPERIENCE

Research Assistant — Machine Learning & Cybersecurity Apr 2026 – Present
Sapienza University of Rome — funded research scholarship (BDR 3/2025), with Prof. Angelo Spognardi and Federica Bianchi. Reimplemented TFE-GNN in PyTorch Geometric; prepared PCAP datasets and designed encrypted-payload ablations and provenance checks to assess model reliability.

Technical Writer & Team Member – Plum Juice Sep 2024 – Present
Published technical documentation of a **32-core Raspberry Pi cluster**, covering MPI/Slurm integration, failures and scaling trade-offs.

Linear Algebra Tutor – Sapienza University of Rome Sep 2025 – Feb 2026
Provided supplementary Linear Algebra classes for the Applied Computer Science & AI BSc programme.

Programming Tutor Nov 2023 – Jan 2024
Improved programming materials using feedback from students in engineering, AI, statistics and economics.

CONTRIBUTIONS TO PUBLIC PROJECTS

Surgical Workflow Analysis – four-person research team Jul 2026

- Reached **92.12% frame accuracy** and **88.36% mean segmental F1** with different models in a 21-configuration PyTorch benchmark on Cholec80; compared temporal quality on **40 held-out videos**.

Literary Translation – two-person NLP project Jun 2025

- Obtained **3.49/5 human-rated quality** with 4-bit Mixtral + LoRA versus **2.21/5** for OPUS-MT across **97 passages**; quantified agreement between human raters and LLM judges.

Fire Detection – computer vision team project May 2023

- Increased measured video-processing throughput from **0.1 to 60 FPS** through NumPy vectorization, enabling real-time processing with adaptive background subtraction and XGBoost.

SELECTED PERSONAL PROJECTS

Samsung Gear S3 Integration – independent Gadgetbridge fork Sep 2026 – Present

- Enabled **notifications, message replies and contact sync** between Android and Tizen. Resolved reconnection and notification-action failures; validated Java/Bluetooth fixes on physical devices and with regression tests.

Encrypted Traffic Classification – TFE-GNN adaptation May – Oct 2026

- Achieved **92.53% mean accuracy across three seeds** on one flow-disjoint split; ran controlled payload/TLS ablations on 1,674 samples and identified capture overlap that prevented independent validation.

Vision Transformer Pruning – mechanistic interpretability May – Jun 2025

- Cut **85% of attention heads** and accelerated CPU inference by **1.85×** using activation patching; retrained the sparse architecture to 55.24% validation accuracy on 58 Tiny ImageNet classes.

SKILLS

Programming: Python, C/C++, Java, Go, SQL, JavaScript, R.
ML / systems: PyTorch, PyTorch Geometric, Hugging Face, NumPy, CUDA, Linux, Git, Docker.
Languages: Italian (C2), English (C2), Spanish (A2), Polish (A2).